Computational Methods for Fast Isosurface and Topological Preimage Extraction on Structured Voxel Grids

Technical Documentation — `LC::Math::MeshProcessor`

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Abstract

This document presents the mathematical formulations and computational algorithms implemented in `MeshProcessor` for high-throughput extraction of scalar isosurfaces and vector-field preimage curves (Pontryagin–Thom constructions) on three-dimensional structured grids. Specifically, we detail: (1) parallel scalar isosurface extraction via 3D Flying Edges with direct buffer emission, (2) branchless orthonormal frame construction on S^2, (3) a direct cell-face 2D barycentric zero-locus solver that avoids intermediate surface-surface intersection filters, (4) spatial hashing and non-shrinking Taubin smoothing for piecewise-linear curves, and (5) parallel-transport (Bishop) frame tube mesh generation with closed-loop holonomy untwisting and flat axial end-caps.

1 Scalar Isosurface Extraction via 3D Flying Edges

Given a three-dimensional structured grid Ω of dimensions $N_x \times N_y \times N_z$ with grid origin $\mathbf{o} = (o_x, o_y, o_z)$ and uniform or rectilinear voxel spacing $\Delta\mathbf{x} = (\Delta x, \Delta y, \Delta z)$, let $s : \Omega \to \mathbb{R}$ denote a discrete scalar field stored in contiguous x-fastest order:

$$\mathrm{idx}(i,j,k) = i + N_x\,(j + N_y k)\,, \quad 0 \le i < N_x,\ 0 \le j < N_y,\ 0 \le k < N_z. \tag{1}$$

Classical Marching Cubes (`vtkMarchingCubes`) evaluates each voxel cell independently and relies on a spatial hash table (`vtkMergePoints`) to weld coincident vertices across shared cell edges, creating a serial synchronization bottleneck. Instead, the pipeline employs the multi-pass **3D Flying Edges** algorithm (`vtkFlyingEdges3D`):

1. **Pass 1 (Edge Classification):** Grid edges parallel to the x-axis are swept in parallel across (j,k) grid lines to determine sign changes relative to the isovalue $s(\mathbf{x}) = s_0$.
2. **Pass 2 (Topological Configuration & Prefix Sum):** Using the x-edge cases of the four edges bounding each (j,k) voxel row, y- and z-edge intersections and triangle counts are tabulated without redundant edge evaluations. A parallel prefix sum allocates exact output vertex and triangle offsets.
3. **Pass 3 (Direct Geometry Generation):** Vertices, central-difference gradients (unit normals), and triangle connectivity are written directly into preallocated contiguous arrays without point-locator lookups.

To eliminate virtual function call overhead during mesh extraction, the underlying single-precision pointers (`float*`) for positions and normals are accessed directly via `vtkFloatArray::FastDownCast`, and cell connectivity is traversed via `vtkCellArrayIterator`.

2 Preimage Curve Formulation and Orthonormal Basis Construction

Let $\mathbf{n} : \Omega \to S^2 \subset \mathbb{R}^3$ be a unit vector field (such as a liquid crystal director field) stored in structure-of-arrays (SoA) layout with contiguous component buffers (n_x, n_y, n_z) of length $V = N_x N_y N_z$. Given a target orientation $\mathbf{n}_0 \in S^2$ (normalized such that $\|\mathbf{n}_0\|_2 = 1$), its preimage is the codimension-2 submanifold:

$$\mathcal{L}(\mathbf{n}_0) = \{\mathbf{x} \in \Omega \mid \mathbf{n}(\mathbf{x}) = \mathbf{n}_0\}. \tag{2}$$

2.1 Branchless Orthonormal Basis on S^2

To convert the vector equality $\mathbf{n}(\mathbf{x}) = \mathbf{n}_0$ into two independent scalar zero-crossing conditions, we construct an orthonormal basis $\{\mathbf{n}_0, \mathbf{b}_1, \mathbf{b}_2\}$ of $\mathbb{R}^3$ such that $\mathbf{b}_1 \perp \mathbf{n}_0$, $\mathbf{b}_2 \perp \mathbf{n}_0$, and $\mathbf{b}_1 \perp \mathbf{b}_2$. Using the branchless Frisvad–Duff construction for $\mathbf{u} = (u_x, u_y, u_z) = \mathbf{n}_0/\|\mathbf{n}_0\|_2$, let $\sigma = \text{copysign}(1, u_z) \in \{-1, +1\}$, and define:

$$a = -\frac{1}{\sigma + u_z}, \qquad b = u_x u_y a. \tag{3}$$

The orthogonal tangent vectors $\mathbf{b}_1, \mathbf{b}_2 \in T_{\mathbf{n}_0} S^2$ are given analytically without singular conditional branches by:

$$\mathbf{b}_1 = \begin{bmatrix} 1 + \sigma u_x^2 a \\ \sigma b \\ -\sigma u_x \end{bmatrix}, \qquad \mathbf{b}_2 = \begin{bmatrix} b \\ \sigma + u_y^2 a \\ -u_y \end{bmatrix}. \tag{4}$$

2.2 Scalar Decomposition and Hemisphere Constraint

Projecting $\mathbf{n}(\mathbf{x})$ onto $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{n}_0\}$ defines three scalar fields:

$$f_1(\mathbf{x}) = \mathbf{n}(\mathbf{x}) \cdot \mathbf{b}_1, \tag{5}$$
$$f_2(\mathbf{x}) = \mathbf{n}(\mathbf{x}) \cdot \mathbf{b}_2, \tag{6}$$
$$\alpha(\mathbf{x}) = \mathbf{n}(\mathbf{x}) \cdot \mathbf{n}_0. \tag{7}$$

The common zero locus $f_1(\mathbf{x}) = 0 \wedge f_2(\mathbf{x}) = 0$ contains both the parallel preimage ($\mathbf{n}(\mathbf{x}) = +\mathbf{n}_0$) and the antiparallel preimage ($\mathbf{n}(\mathbf{x}) = -\mathbf{n}_0$), as well as numerical interpolation artifacts across steep defect cores where $\|\mathbf{n}(\mathbf{x})\|$ degrades. Imposing the polar alignment threshold $\alpha(\mathbf{x}) \geq \tau_{\text{align}}$ (typically $\tau_{\text{align}} \in (0, 1)$, e.g., 0.5) isolates the true $+\mathbf{n}_0$ preimage:

$$\mathcal{L}(\mathbf{n}_0) = \{\mathbf{x} \in \Omega \mid f_1(\mathbf{x}) = 0,\ f_2(\mathbf{x}) = 0,\ \alpha(\mathbf{x}) \geq \tau_{\text{align}}\}. \tag{8}$$

3 Direct 2D Barycentric Zero-Locus Solver on Voxel Faces

Computing $\mathcal{L}(\mathbf{n}_0)$ by extracting two global 3D isosurfaces $\mathcal{S}_1 = f_1^{-1}(0)$ and $\mathcal{S}_2 = f_2^{-1}(0)$ and intersecting them via an octree triangle-triangle intersection filter (`vtkIntersectionPolyDataFilter`) incurs $O(V + M \log M)$ overhead and suffers from topological failures when clipping introduces degenerate triangles. Instead, we solve for the zero locus directly on the faces of each voxel cell.

3.1 Early Voxel Rejection

For each voxel cell (i, j, k) with 8 corner indices $c \in \{0, \ldots, 7\}$, we evaluate $f_1^{(c)}, f_2^{(c)}, \alpha^{(c)}$ at the eight corners. By the convex hull property of piecewise-linear interpolation, the cell cannot contain a valid preimage segment and is immediately skipped unless all three conditions hold across its 8 corners:

1. $\max_c \alpha^{(c)} \geq \tau_{\text{align}}$ (at least one corner lies in the target alignment cone),

2. $\min_c f_1^{(c)} \leq 0 \ \wedge \ \max_c f_1^{(c)} \geq 0$ (f_1 changes sign across the cell),

3. $\min_c f_2^{(c)} \leq 0 \ \wedge \ \max_c f_2^{(c)} \geq 0$ (f_2 changes sign across the cell).

3.2 Consistent Face Triangulation and Cramer's Rule Solver

Each of the 6 rectangular faces of the voxel cube (i, j, k) is split into two triangles $(A, B, C) = (\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2)$ and $(\mathbf{p}_0, \mathbf{p}_2, \mathbf{p}_3)$ along a canonical diagonal ordered by local vertex indices:

Voxel Face	**Quad Corner Indices** (v_0, v_1, v_2, v_3)	**Shared Triangles**
$-X$ / $+X$	$(0, 2, 6, 4)$ / $(1, 3, 7, 5)$	(v_0, v_1, v_2) and (v_0, v_2, v_3)
$-Y$ / $+Y$	$(0, 1, 5, 4)$ / $(2, 3, 7, 6)$	(v_0, v_1, v_2) and (v_0, v_2, v_3)
$-Z$ / $+Z$	$(0, 1, 3, 2)$ / $(4, 5, 7, 6)$	(v_0, v_1, v_2) and (v_0, v_2, v_3)

Table 1: Canonical face quad indexing ensuring identical diagonal splits between adjacent voxels.

Because opposite faces $(-X, +X)$, $(-Y, +Y)$, and $(-Z, +Z)$ use identical local vertex orderings, two adjacent voxels share the exact same two triangles on their common boundary face.

On any face triangle (A, B, C) with vertex positions $\mathbf{p}_A, \mathbf{p}_B, \mathbf{p}_C \in \mathbb{R}^3$, any point $\mathbf{x}$ is parameterized by barycentric coordinates (w_A, u_B, v_C) with $w_A = 1 - u_B - v_C$:

$$\mathbf{x}(u_B, v_C) = w_A \mathbf{p}_A + u_B \mathbf{p}_B + v_C \mathbf{p}_C. \tag{9}$$

Affine interpolation of f_1 and f_2 over the triangle yields the 2×2 linear system $f_1(u_B, v_C) = 0$ and $f_2(u_B, v_C) = 0$:

$$\begin{bmatrix} \Delta_{1B} & \Delta_{1C} \\ \Delta_{2B} & \Delta_{2C} \end{bmatrix} \begin{bmatrix} u_B \\ v_C \end{bmatrix} = \begin{bmatrix} -f_{1A} \\ -f_{2A} \end{bmatrix}, \tag{10}$$

where $\Delta_{1B} = f_{1B} - f_{1A}$, $\Delta_{1C} = f_{1C} - f_{1A}$, $\Delta_{2B} = f_{2B} - f_{2A}$, and $\Delta_{2C} = f_{2C} - f_{2A}$. Let the determinant be $D = \Delta_{1B}\Delta_{2C} - \Delta_{1C}\Delta_{2B}$. If $|D| > 10^{-12}$, Cramer's rule gives the unique zero crossing:

$$u_B = \frac{-f_{1A}\Delta_{2C} + f_{2A}\Delta_{1C}}{D}, \qquad v_C = \frac{-\Delta_{1B} f_{2A} + \Delta_{2B} f_{1A}}{D}, \qquad w_A = 1 - u_B - v_C. \tag{11}$$

The intersection is valid if and only if the point lies inside the triangle ($u_B \geq -\epsilon, v_C \geq -\epsilon, w_A \geq -\epsilon$ with $\epsilon = 10^{-5}$) and the interpolated alignment satisfies:

$$\alpha(u_B, v_C) = w_A \alpha_A + u_B \alpha_B + v_C \alpha_C \geq \tau_{\text{align}}. \tag{12}$$

Collecting the distinct face intersections within each active voxel yields a pair of entry and exit points $(\mathbf{q}_0, \mathbf{q}_1)$, defining an elementary line segment of the preimage curve.

4 Polyline Reconstruction and Non-Shrinking Taubin Smoothing

4.1 Spatial Hashing and Graph Stripping

To assemble the unorganized cell segments into continuous polylines in $O(M)$ time, segment endpoints are quantized into spatial hash buckets of width $h_{\text{weld}} = 0.35\, h_{\min}$, where $h_{\min} = \min(\Delta x, \Delta y, \Delta z)$:

$$\mathbf{g}(\mathbf{p}) = (\lfloor p_x / h_{\text{weld}} \rfloor, \ \lfloor p_y / h_{\text{weld}} \rfloor, \ \lfloor p_z / h_{\text{weld}} \rfloor) \in \mathbb{Z}^3, \tag{13}$$

hashed via large prime XOR mixing. Each incoming vertex queries its $3 \times 3 \times 3$ neighborhood of grid buckets and welds to the closest existing vertex within Euclidean distance h_{weld}.

An undirected adjacency graph is constructed over the welded vertices. Contiguous polyline chains are traced by initiating depth-first walks from non-manifold or degree-1 endpoints first, followed by unvisited degree-2 closed cycles. Open chains whose endpoints lie within $4\,h_{\min}$ are greedily bridged, and chains whose start and end vertices lie within $4\,h_{\min}$ are flagged as closed loops.

4.2 Subdivision and 1D Taubin $\lambda|\mu$ Smoothing

To eliminate staircase artifacts inherited from the voxel grid without shrinking closed preimage rings, each polyline chain $\Gamma = \{\mathbf{p}_0, \ldots, \mathbf{p}_{M-1}\}$ undergoes one midpoint subdivision pass followed by $N_{\text{iter}} = 25$ iterations of 1D Taubin smoothing.

Define the discrete 1D umbrella Laplacian operator $\mathcal{U}(\mathbf{p}_i)$ at interior or periodic vertex i:

$$\mathcal{U}(\mathbf{p}_i) = \frac{1}{2}\left(\mathbf{p}_{i-1} + \mathbf{p}_{i+1}\right) - \mathbf{p}_i, \tag{14}$$

with indices evaluated modulo M for closed curves and pinned ($\mathcal{U}(\mathbf{p}_0) = \mathcal{U}(\mathbf{p}_{M-1}) = \mathbf{0}$) for open curves. Each Taubin iteration applies a forward shrinking pass with weight $\lambda = 0.50$ followed by a backward inflation pass with weight $\mu = -0.53$:

$$\mathbf{p}_i^{(k+1/2)} = \mathbf{p}_i^{(k)} + \lambda\mathcal{U}\left(\mathbf{p}_i^{(k)}\right), \tag{15}$$

$$\mathbf{p}_i^{(k+1)} = \mathbf{p}_i^{(k+1/2)} + \mu\mathcal{U}\left(\mathbf{p}_i^{(k+1/2)}\right). \tag{16}$$

Because $\lambda + \mu < 0$, the transfer function $(1 - \lambda\omega)(1 - \mu\omega)$ attenuates high-frequency voxel staircase noise ($\omega \approx 2$) while preserving low-frequency curve curvature and enclosed loop area ($\omega \to 0$).

5 Parallel-Transport Tube Generation and End-Capping

5.1 Bishop Frame Propagation and Holonomy Correction

Given a smoothed polyline $\Gamma = \{\mathbf{p}_0, \ldots, \mathbf{p}_{M-1}\}$, central differences yield the unit tangent vector $\mathbf{T}_i$ at each vertex:

$$\mathbf{T}_i = \frac{\mathbf{p}_{\text{next}(i)} - \mathbf{p}_{\text{prev}(i)}}{\|\mathbf{p}_{\text{next}(i)} - \mathbf{p}_{\text{prev}(i)}\|_2}. \tag{17}$$

Unlike the Frenet–Serret frame, which becomes undefined when curvature vanishes and flips discontinuously across inflection points, we construct a discrete **parallel-transport (Bishop) frame** $\{\mathbf{T}_i, \mathbf{N}_i, \mathbf{B}_i\}$. Starting from an initial orthonormal pair $(\mathbf{N}_0, \mathbf{B}_0) \perp \mathbf{T}_0$, each subsequent normal $\mathbf{N}_i$ is obtained by projecting $\mathbf{N}_{i-1}$ orthogonal to $\mathbf{T}_i$:

$$\tilde{\mathbf{N}}_i = \mathbf{N}_{i-1} - (\mathbf{N}_{i-1} \cdot \mathbf{T}_i)\,\mathbf{T}_i, \qquad \mathbf{N}_i = \frac{\tilde{\mathbf{N}}_i}{\|\tilde{\mathbf{N}}_i\|_2}, \qquad \mathbf{B}_i = \mathbf{T}_i \times \mathbf{N}_i. \tag{18}$$

For closed loops, parallel transport around the full cycle accumulates a geometric Berry/holonomy angle $\Delta\theta \in [-\pi, \pi]$ between the transported final normal $\mathbf{N}_{\text{trans}}$ (projected onto $T_{\mathbf{p}_0}\Gamma^{\perp}$) and the starting normal $\mathbf{N}_0$:

$$\Delta\theta = \operatorname{atan2}(\mathbf{N}_{\text{trans}} \cdot \mathbf{B}_0,\ \mathbf{N}_{\text{trans}} \cdot \mathbf{N}_0). \tag{19}$$

To ensure C^1-continuous tube connectivity across the closure seam, we counter-rotate each local frame $(\mathbf{N}_i, \mathbf{B}_i)$ in its normal plane by $\phi_i = -\Delta\theta\frac{i}{M}$:

$$\begin{bmatrix} \mathbf{N}_i^{\text{corr}} \\ \mathbf{B}_i^{\text{corr}} \end{bmatrix} = \begin{bmatrix} \cos\phi_i & \sin\phi_i \\ -\sin\phi_i & \cos\phi_i \end{bmatrix} \begin{bmatrix} \mathbf{N}_i \\ \mathbf{B}_i \end{bmatrix}. \tag{20}$$

5.2 Tubular Surface and Analytical Normal Emission

For a tube of radius R with S circumferential sides, let $\theta_s = 2\pi s/S$ for $s \in \{0, \ldots, S-1\}$. The outward unit radial normal $\mathbf{r}_{i,s}$ and surface vertex position $\mathbf{v}_{i,s}$ at ring i and sector s are emitted directly:

$$\mathbf{r}_{i,s} = \cos\theta_s \, \mathbf{N}_i^{\text{corr}} + \sin\theta_s \, \mathbf{B}_i^{\text{corr}}, \qquad \mathbf{v}_{i,s} = \mathbf{p}_i + R\,\mathbf{r}_{i,s}. \tag{21}$$

Each quad between adjacent rings i and $i+1$ and sectors s and $(s+1) \bmod S$ is tessellated into two outward-oriented triangles (i_{00}, i_{10}, i_{11}) and (i_{00}, i_{11}, i_{01}).

5.3 Dedicated Flat Axial End-Caps for Open Curves

For open preimage curves, sharing vertices between the cylindrical wall and the end-caps would average radial and axial normals, causing shading artifacts at the rims. Instead, we emit $S+1$ dedicated vertices at the start ($i = 0$, outward normal $-\mathbf{T}_0$) and $S+1$ dedicated vertices at the end ($i = M-1$, outward normal $+\mathbf{T}_{M-1}$), consisting of the centerline endpoint $\mathbf{p}_{\text{end}}$ and a duplicate rim ring $\mathbf{p}_{\text{end}} + R\,\mathbf{r}_{\text{end},s}$. Connecting the center vertex to adjacent rim vertices via triangle fans $(c_{\text{start}}, s+1, s)$ and $(c_{\text{end}}, s, s+1)$ preserves sharp geometric creases at both boundaries.